1. PLD's and ASICS compare in terms of design flexibility and performance and application.

2H00030520

A.

ASICS	PLD'S
→ Fixed after fabrica-tion no changes are possible	→ High flexible can be performed (or) programmed many times
→ very high performan-ce with optimized pow-er and speed	→ Moderate performanc-e, lower speed than ASICS.
→ High non-recurring engineering cost but cheaper per uni-t at large volume	→ Lower upfront cost, idea for low to mid volume
→ Longer due to fabrication cycle	→ Faster since pro-gramming is immedi-ate
→ Best for mass produced custom systems	→ Best for prototypi-ng, small systems, reconfigurable logics.

2. Describe the requirements of the embedded System design process with a neat diagram and suitable example.

→ clear functional and non-functional requirements

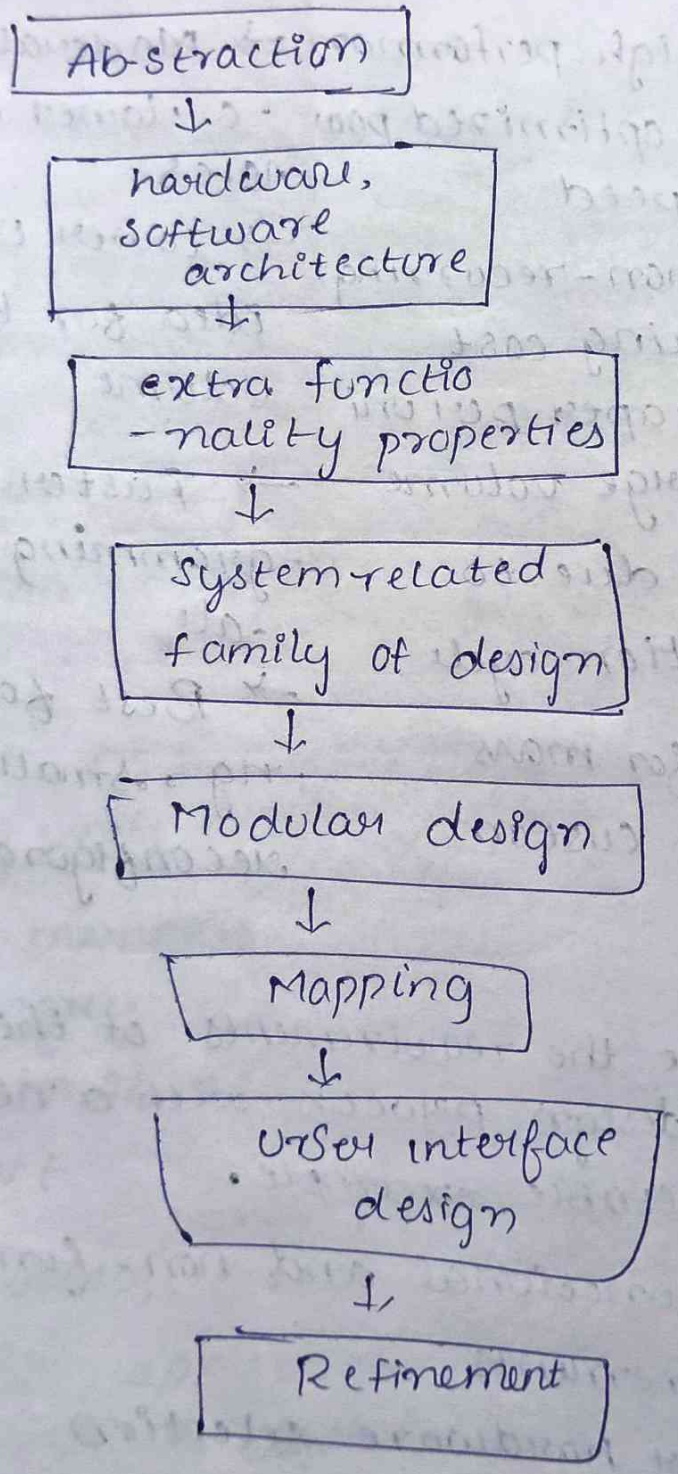
→ proper hardware selection like processor, memomry, input and output devices.

→ software tools for development like firmware, RTOS, etc.. 2400030520

→ Integration and testing

→ optimization for cost, power and performance

Diagram:-



3. Demonstrate the role of architecture design in the embedded system design process by using suitable examples. 2400030520

→ Achitecture design specifies the overall structure of the embedded system, including processor, memory, peripherals.

→ It decides which functions are implemented in hardware and which are implemented in software to achieve optimal performance and cost

→ Architecture ensures that real-time constraints, latency and throughput requirements are met

→ proper allocation of cpu time, memory, power i/o resources is planned during architecture design.

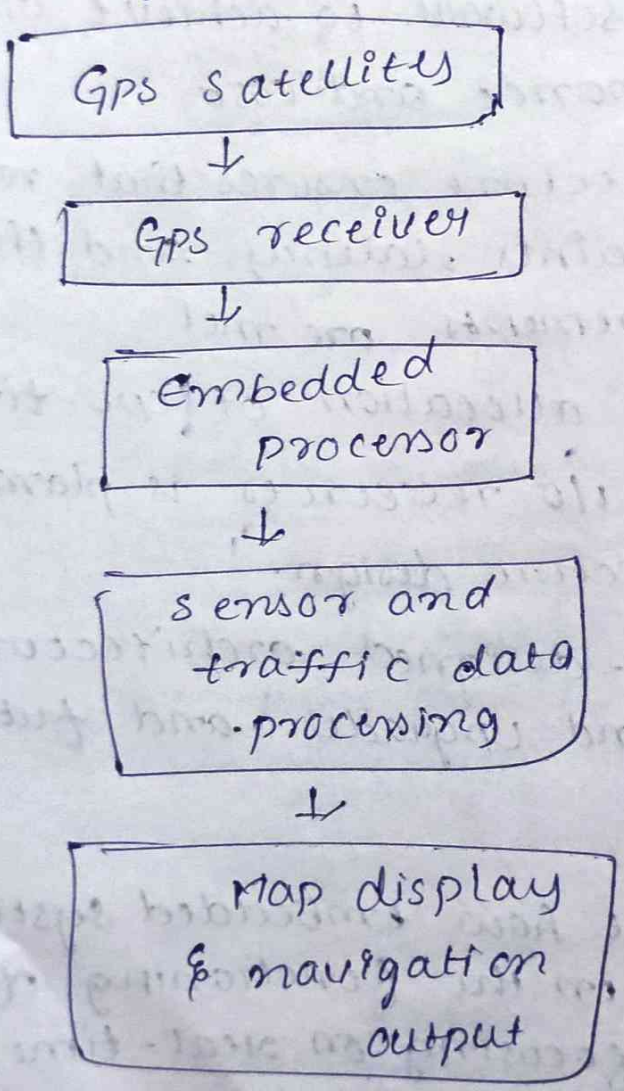
→ A well-designed architecture allows easy and upgrades and future enhancements

4. Illustrate how embedded system are applied in the functioning of google maps, focusing on real-time data processing and GPS integeration.

Embedded system plays a vital role in the functioning of google maps by enabling GPS integration and real-time data processing.

2400030520

→ GPS integeration
→ real-time data processing
→ sensor data fusion
→ Traffic and route updates
→ user interface and navigation



5. Discuss the role and importance of embedded system in modern healthcare provide specific examples

2H00030520

→ patient monitoring
→ Accurate Diagnosis
→ Life saving devices
→ portable and wearable healthcare devices
→ Real time data processing

Ex :- * ECG and ICU monitor
* pacemakers, MRI, CT scan etc..